

Included tools and utility programs

The toolkit comes with a large range of built-in tools and programs for the developer. Let us take a look at some of the most commonly used tools.

uhd_fft. By using this simple spectrum analyser tool, we can display the spectrum at a given frequency by connecting a universal software radio peripheral hardware driver (UHD) device. The tool can also be used for a waterfall plot or as an oscilloscope.

uhd_rx_cfile. The function of this tool is to record an I/Q sample stream using a connected UHD device. Samples are written to a file and can be analysed offline at a later time, using either GNU Radio or other tools, such as Octave or MATLAB.

uhd_rx_nogui. This tool receives and listens to incoming signals on your audio device. It can demodulate AM and FM signals.

uhd_siggen_gui.py. This is a simple signal generator application that can create the most common signals, such as sine, sweep, square and noise.

gr_plot*. This tool contains an entire suite of apps that can display pre-recorded samples saved to a file. The user can plot spectra, PSD and time-domain representations of these signals.

We can call these applications from the command line in Linux. A help switch -h is included with all these applications.

flexible enough for your application. In this case, you may add your own blocks. Building blocks in C++ language ensure that speed and performance are not compromised. But if you want to quickly prototype something without arguing with C++, you can opt for Python.

GNU Radio blocks are also avail-

able in Python, even if these are written in C++. This is made possible with the help of simple wrapper and interference generator (SWIG). Glue codes are automatically generated with the help of this functionality. This is done in order to help users code everything themselves and to enable them to enjoy the power and

flexibility of Python and its libraries for Python-centric processing of signals, or making use of their favourite widget library to create any GUI.

Bringing the code closest to the antenna. The main idea behind the GNU Radio project, as described by project founder Eric Blossom, is to bring the code as close to the antenna as possible and to turn hardware problems into software problems.

People around the world make use of GNU Radio along with readily-available low-cost external RF hardware to develop real-world SDRs. These are also run without hardware in simulation-like environments. With developments in communication techniques, like orthogonal frequency division multiplexing, the number of users for the tool has been increasing exponentially. Hopefully, in the near future we can expect the vision behind the project to be fulfilled. ●

The author is an electronics enthusiast from Kerala

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